Question 1: Dynamic Programming — To be answered in **(10) Minutes** — [ ] / 5 Marks

Use **Dynamic Programming** to find the length and the actual sequence of the **Longest Common Subsequence (LCS)** for the following two strings:

String X: ABA

String Y: ACCBA

1. Fill the table below.

$A=X_i$ \ $B=Y_j$		0	1	2	3	4	5
		""	A	C	C	B	A
0	""						
1	A						
2	B						
3	A						

2. The LCS is _____

3. The length of the LCS is _____

Student Name (in English): _____ Student ID: _____

Question 2: Greedy Algorithms To be answered in **(20) Minutes** [ ] / 5 Marks

A. Given the **fractional knapsack** capacity **W = 40**, and items with benefits and weights: **(2 marks)**

- **Item A:** benefit= 60, weight = 30
- **Item B:** benefit= 60, weight = 20
- **Item C:** benefit= 120, weight = 30

1. Use a **greedy algorithm** to select which items to include in the knapsack that maximize the total benefit. Show the **details**.

2. What is the knapsack **total benefit** of the above answer? _____

B. Consider the following set of **five activities (x, y, z, w, and t)** with their corresponding start and finish times. Select the maximum number of non-overlapping activities using a **greedy algorithm**. Show all the **details**. **(3 marks)**

Activity	x	y	z	w	t
Start time	3	1	5	8	0
Finish time	5	4	7	9	6

Question 1:

To be answered in (30) Minutes [ ] / 10 Marks

Given a set of coin denominations {1, 5, 10, 50, 100, 500} and an infinite supply of each coin denomination.

1. Apply a **greedy algorithm** to find the minimum number of coins needed to make change for the amount **n = 12**. Show all the **details**. **[2 marks]**

2. Apply a **dynamic programming** algorithm to find the minimum number of coins needed to make change for the amount **n = 12**. Show all the **details** including filling the table below. **[4 marks]**

0	1	2	3	4	5	6	7	8	9	10	11	12

3. Given an infinite supply of each coin denominations **{1, 5, 10, 50, 100, 500}**, which algorithm is **better** to use? **[1 mark]**
   a. Brute Force
   b. Greedy
   c. Dynamic programming

4. If the set of coin denominations is changed to **{1, 6, 10}**, which coins will be selected by a greedy algorithm to make change for **n = 12**? (No details needed). **[1 mark]**

5. If the set of coin denominations is changed to **{1, 6, 10}**, which coins will be selected by a **dynamic programming algorithm** to make change for **n=12**? (No details needed). **[1 mark]**

6. If the set of coin denominations is changed to **{1, 6, 10}**, which algorithm gives the **optimal** solution? **[1 mark]**
   a. Brute Force
   b. Greedy
   c. Dynamic programming